CSE4062
Introduction to Data Science and Analytics
Spring 2023

DELIVERY 3

IT Support Tickets

Descriptive Analytics

MEMBERS

1. Introduction

We tested the dataset with various clustering algorithms (DBSCAN, K-Means, Agglomerative Clustering, Apriori). We tested unsupervised clustering algorithms on numerical columns only. Our target attribute is *URGENCY*.

- Complition Time
- Total_Assignee
- Total_Worklog_Assginee
- Total_Log_Hours_Assignee
- COMMENTOR_COUNT
- COMMENT_COUNT
- WAITING_HOURS_AT_CUSTOMER

2. Results

a. Apriori Results

Apriori algorithm executed on these columns only.

- IMPACT_Operation Continue With Workaround
- PRIORITY_Major
- IMPACT_Some Processes Stop on Production
- IMPACT_Operation Continue With Workaround
- PRIORITY_Minor

Findings

1. The first association rule shows that when there is an impact on some processes stopping production, there is a 44.0% chance of having a minor priority. This rule has a lift value of 1.153, suggesting a slightly positive correlation between the antecedent and consequent.
2. The second association rule indicates that when the priority is minor, there is a 73.8% chance of impacting some processes to stop production. This rule also has a lift value of 1.153, indicating a slight positive correlation between the antecedent and consequent.
3. The third association rule suggests that when the priority is minor, there is an 18.4% chance of continuing operations with a workaround. This rule has a lift value of 1.112, indicating a slightly positive correlation.

4. The fourth association rule states that when operations continue with a workaround, there is a 42.4% chance of having a minor priority. The lift value of 1.112 suggests a slight positive correlation between the antecedent and consequent.
5. The fifth association rule shows that when operations continue with a workaround, there is a 52.2% chance of having a major priority. The lift value of 1.036 suggests a weak positive correlation between the antecedent and consequent.

Results

	antecedents
4	(IMPACT_Some Processes Stop on Production)
5	(PRIORITY_Minor)
2	(PRIORITY_Minor)
3	(IMPACT_Operation Continue With Workaround)
0	(IMPACT_Operation Continue With Workaround)
1	(PRIORITY_Major)

	consequents	antecedent support
4	(PRIORITY_Minor)	0.640080
5	(IMPACT_Some Processes Stop on Production)	0.381633
2	(IMPACT_Operation Continue With Workaround)	0.381633
3	(PRIORITY_Minor)	0.165301
0	(PRIORITY_Major)	0.165301
1	(IMPACT_Operation Continue With Workaround)	0.503225

	consequent support	support	confidence	lift	leverage	conviction
4	0.381633	0.281728	0.440145	1.153320	0.037452	1.104512
5	0.640080	0.281728	0.738217	1.153320	0.037452	1.374879
2	0.165301	0.070149	0.183814	1.111997	0.007065	1.022682
3	0.381633	0.070149	0.424374	1.111997	0.007065	1.074252
0	0.503225	0.086209	0.521530	1.036375	0.003026	1.038257
1	0.165301	0.086209	0.171313	1.036375	0.003026	1.007256

	zhangs_metric
4	0.369354
5	0.214982
2	0.162875
3	0.120662
0	0.042049
1	0.070652

Clustering Results

We created 2 length combinations of all numerical columns listed below and run DBSCAN, K-Means and Agglomerative Clustering algorithms. We couldn't find any interesting pattern

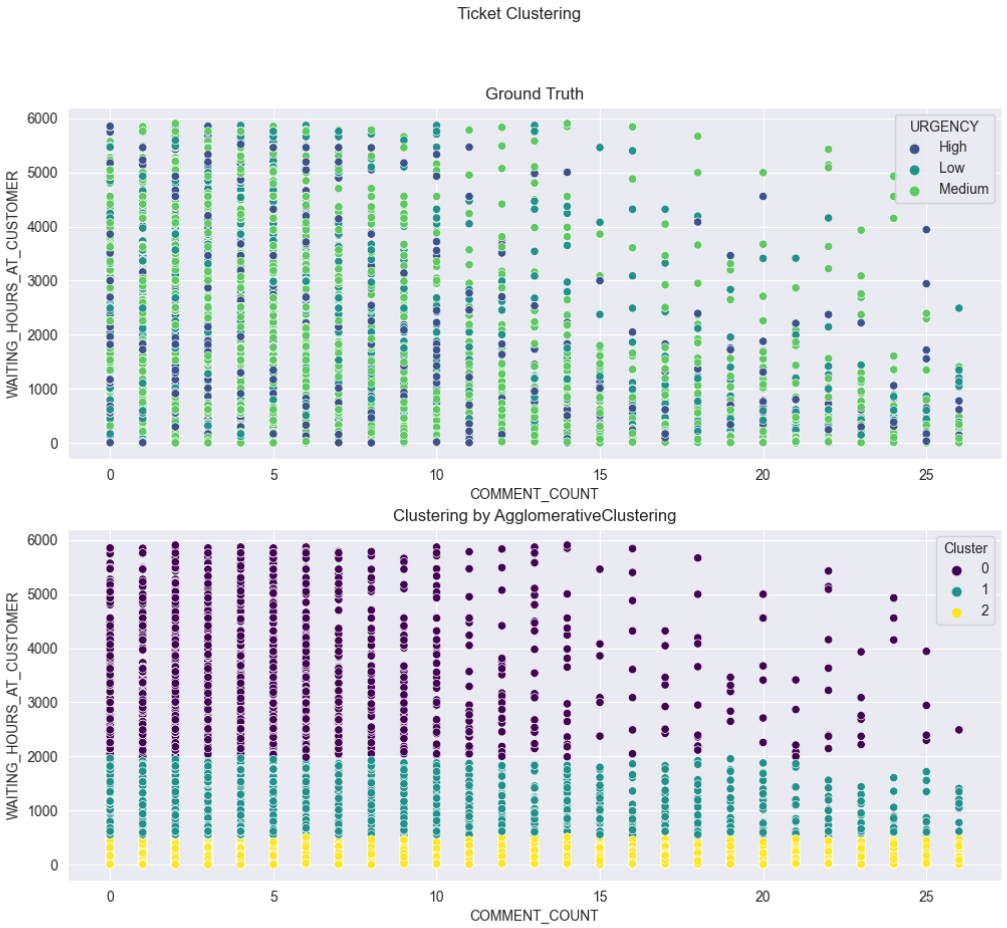
by using these algorithms. We obtained 63 different results with graphs. You can see some of them in the appendix section.

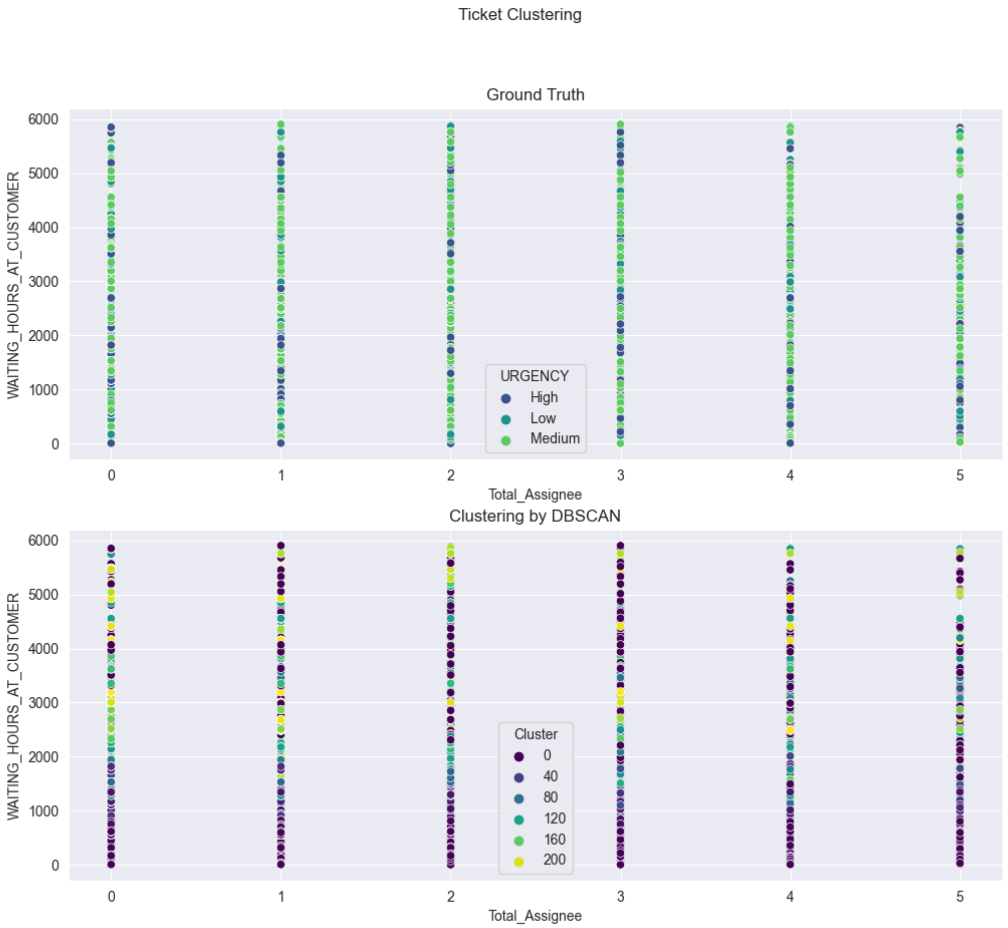
1. Complition Time
2. Total_Assignee
3. Total_Worklog_Assginee
4. Total_Log_Hours_Assignee
5. COMMENTOR_COUNT
6. COMMENT_COUNT
7. WAITING_HOURS_AT_CUSTOMER

3. Conclusion

The Apriori results indicate several associations between different variables. Specifically, there is a correlation between an impact on some processes stopping production and having a minor priority, as well as between a minor priority and impacting processes. Additionally, there is a connection between continuing operations with a workaround and having a minor or major priority. These findings provide insights into the relationships and dependencies among these variables.

4. Appendix


Ticket Clustering
Ground Truth
URGENCY
High
Low
Medium
WAITING_HOURS_AT_CUSTOMER
COMMENT_COUNT
Clustering by AgglomerativeClustering
Cluster
0
1
2
WAITING_HOURS_AT_CUSTOMER
COMMENT_COUNT


Ticket Clustering
Ground Truth
WAITING_HOURS_AT_CUSTOMER
6000
5000
4000
3000
2000
1000
0
URGENCY
High
Low
Medium
0
1
2
3
4
5
Total_Assignee
Clustering by DBSCAN
WAITING_HOURS_AT_CUSTOMER
Cluster
0
40
80
120
160
200
Total_Assignee


Ticket Clustering
Ground Truth
WAITING_HOURS_AT_CUSTOMER
Complition Time
URGENCY
High
Low
Medium
Clustering by KMeans
WAITING_HOURS_AT_CUSTOMER
Complition Time
Cluster
0
1
2